

MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Height and Distance

(English Medium)

Copyright Statement

All rights including trademark and copyrights and rights of translation etc. reserved and vested exclusively with ALLEN Career Institute Private Limited. (ALLEN)

No part of this work may be copied, reproduced, adapted, abridged or translated, transcribed, transmitted, stored or distributed in any form retrieval system, computer system, photographic or other system or transmitted in any form or by any means whether electronic, magnetic, chemical or manual, mechanical, digital, optical, photocopying, recording or otherwise, or stored in any retrieval system of any nature without the written permission of the Allen Career Institute Private Limited. Any breach will entail legal action and prosecution without further notice.

This work is sold/distributed by Allen Career Institute Private Limited subject to the condition and undertaking given by the student that all proprietary rights (under the Trademark Act, 1999 and Copyright Act, 1957) of the work shall be exclusively belong to ALLEN Career Institute Private Limited. Neither the Study Materials and/or Test Series and/or the contents nor any part thereof i.e. work shall be reproduced, modify, re-publish, sub-license, upload on website, broadcast, post, transmit, disseminate, distribute, sell in market, stored in a retrieval system or transmitted in any form or by any means for reproducing or making multiple copies of it.

Any person who does any unauthorised act in relation to this work may be liable to criminal prosecution and civil claims for damages. Any violation or infringement of the propriety rights of Allen shall be punishable under Section- 29 & 52 of the Trademark Act, 1999 and under Section- 51, 58 & 63 of the Copyright Act, 1957 and any other Act applicable in India. All disputes are subjected to the exclusive jurisdiction of courts, tribunals and forums at Kota, Rajasthan only.

Note:- This publication is meant for educational and learning purposes. All reasonable care and diligence have been taken while editing and printing this publication. ALLEN Career Institute Private Limited shall not hold any responsibility for any error that may have inadvertently crept in. ALLEN Career Institute Private Limited is not responsible for the consequences of any action taken on the basis of this publication.

Ex.3 The angle of elevation of the tower observed from each of the three point A,B,C on the ground, forming a triangle is the same angle α . If R is the circum - radius of the triangle ABC, then the height of the tower is -

- (1) $R \sin \alpha$ (2) $R \cos \alpha$ (3) $R \cot \alpha$ (4) $R \tan \alpha$

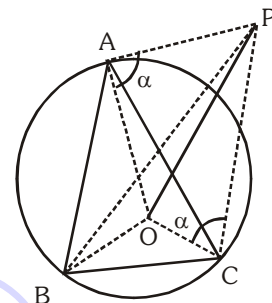
Sol.(4) The tower makes equal angles at the vertices of the triangle, therefore foot of the tower is at the circumcentre.

From ΔOCP , OP is perpendicular to OC.

$$\angle OCP = \alpha$$

$$\text{so } \tan \alpha = \frac{OP}{OA} \Rightarrow OP = OA \tan \alpha$$

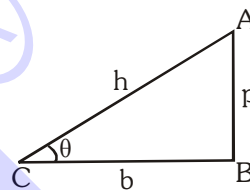
$$OP = R \tan \alpha$$



3. SOME USEFUL RESULTS :

- In a triangle ABC,

$$\sin \theta = \frac{p}{h}, \quad \cos \theta = \frac{b}{h}, \quad \tan \theta = \frac{p}{b}$$

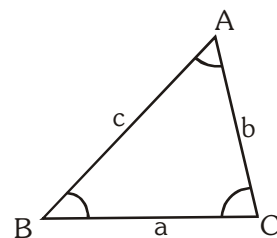


- In any triangle ABC,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \quad [\text{By sine rule}]$$

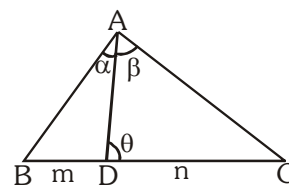
or cosine formula

$$\text{i.e. } \cos A = \frac{b^2 + c^2 - a^2}{2bc}; \cos B = \frac{a^2 + c^2 - b^2}{2ac}, \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



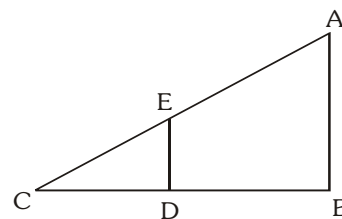
- In any triangle ABC

if $BD : DC = m : n$ and $\angle BAD = \alpha$
 $\angle CAD = \beta$ and $\angle ADC = \theta$,
 then $(m+n) \cot \theta = m \cot \alpha + n \cot \beta$



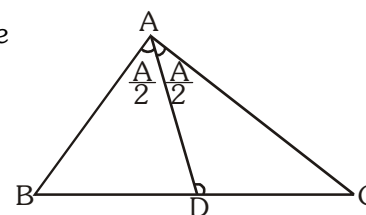
- In a triangle ABC, if $DE \parallel AB$

$$\text{then, } \frac{AB}{DE} = \frac{BC}{DC}$$



- In a triangle the internal bisector of an angle divides the opposite side in the ratio of the arms of the angle

$$\therefore \frac{BD}{DC} = \frac{AB}{AC}$$



- In an isosceles triangle the median is perpendicular to the base

SOLVED EXAMPLES

Ex.1 A tower subtends an angle of 30° at a point on the same level as its foot, and at a second point h m above the first, the depression of the foot of tower is 60° . The height of the tower is.

- (1) h m (2) $3h$ m (3) $\sqrt{3} h$ m (4) $\frac{h}{3}$ m.

Sol.(4) Let OP be the tower of height x , A the point on the same level as the foot O of the tower and B be the point h m above A (see Fig.) Then $\angle AOB = 60^\circ$ and $\angle PAO = 30^\circ$. From right-angled triangle AOP , we have

$$OA = x \cot 30^\circ$$

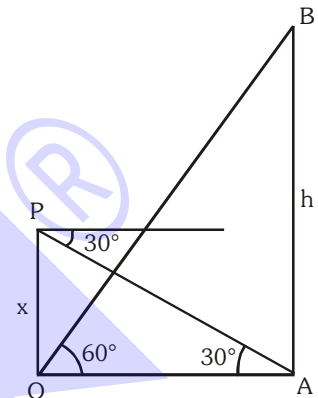
and from right-angled triangle OAB , we have

$$OA = h \cot 60^\circ$$

Therefore, from (1) and (2), we get

$$x \cot 30^\circ = h \cot 60^\circ$$

$$\sqrt{3} x = \frac{1}{\sqrt{3}} h \Rightarrow x = \frac{1}{3} h$$



Ex.2 At a point on level ground, the angle of elevation of a vertical tower is found to be such that its tangent is $\frac{5}{12}$.

On walking 192 metres towards the tower, the tangent of the angle of elevation is $\frac{3}{4}$. Find the height of the tower.

Sol. Let AB be the tower and let the angle of elevation of its top at C be α . Let D be a point at a distance of 192 metres from C such that the angle of elevation of the top of the tower at D be β .

Let h be the height of the tower and $AD = x$,

It is given that $\tan \alpha = \frac{5}{12}$ and $\tan \beta = \frac{3}{4}$.

In $\triangle ABC$, we have

$$\tan \alpha = \frac{AB}{AC} \Rightarrow \tan \alpha = \frac{h}{192 + x} \Rightarrow \frac{5}{12} = \frac{h}{192 + x} \quad \dots (i)$$

In $\triangle ABD$, we have

$$\tan \beta = \frac{AB}{AD} \Rightarrow \tan \beta = \frac{h}{x} \Rightarrow \frac{3}{4} = \frac{h}{x} \quad \dots (ii)$$

We have to find h . This means that we have to eliminate x from (i) and (ii).

From (ii), we have $3x = 4h \Rightarrow x = \frac{4h}{3}$

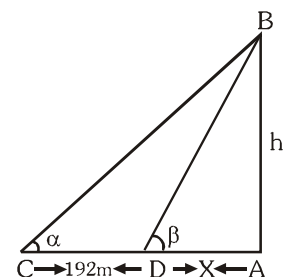
Substituting this value of x in (i), we get

$$\frac{5}{12} = \frac{h}{192 + 4h/3} \Rightarrow 5 \left(192 + \frac{4h}{3} \right) = 12h$$

$$\Rightarrow 5(576 + 4h) = 36h \Rightarrow 2880 + 20h = 36h$$

$$\Rightarrow 16h = 2880 \Rightarrow h = \frac{2880}{16} = 180$$

Hence, height of tower = 180 metres.



Ex.3 Let α be the solution of $16^{\sin^2 \theta} + 16^{\cos^2 \theta} = 10$ in $(0, \pi/4)$. If the shadow of a vertical pole is $\frac{1}{\sqrt{3}}$ of its height, then the altitude of the sun is-

- (1) α (2) $\frac{\alpha}{2}$ (3) 2α (4) $\frac{\alpha}{3}$

Sol. We have $16^{\sin^2 \theta} + 16^{\cos^2 \theta} = 10$

$$\Rightarrow 16^{\sin^2 \theta} + 16^{1-\sin^2 \theta} = 10 \Rightarrow x + \frac{16}{x} = 10, \text{ where } x = 16^{\sin^2 \theta}$$

$$\Rightarrow x = 2, 8 \Rightarrow 16^{\sin^2 \theta} = 2, 8$$

$$\Rightarrow 2^{4\sin^2 \theta} = 2, 2^3$$

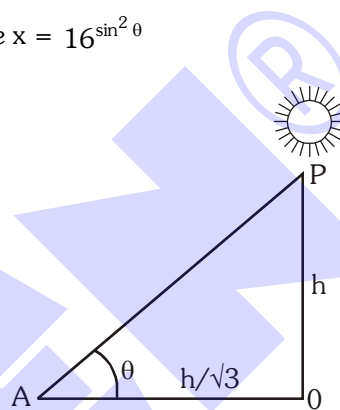
$$\Rightarrow 4\sin^2 \theta = 2, 3$$

$$\Rightarrow \sin^2 \theta = \frac{1}{2}, \left(\frac{\sqrt{3}}{2}\right)^2 \Rightarrow \theta = \frac{\pi}{6}, \frac{\pi}{3}$$

$$\therefore \alpha = \frac{\pi}{6}$$

Let the altitude of the sun be θ . Then,

$$\tan \theta = \frac{h}{\frac{h}{\sqrt{3}}} \Rightarrow \tan \theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3} \Rightarrow \theta = 2\alpha$$



Ex.4 A vertical lamp-post of height 9 metres stands at the corner of a rectangular field. The angle of elevation of its top from the farthest corner is 30° , while from another corner it is 45° . The area of the field is-

- (1) $81\sqrt{2} \text{ m}^2$ (2) $9\sqrt{2} \text{ m}^2$ (3) $81\sqrt{3} \text{ m}^2$ (4) $9\sqrt{3} \text{ m}^2$

Sol. Let AP be the lamp-post of 9 m standing at corner A of the rectangular field ABCD.

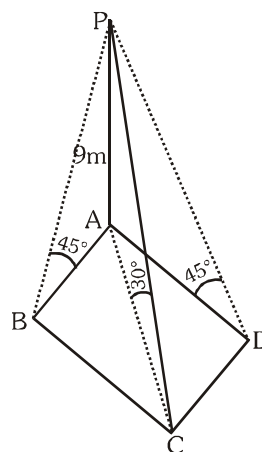
In Δ 's BAP and CAP, we have

$$\tan 45^\circ = \frac{PA}{BA} \text{ and } \tan 30^\circ = \frac{PA}{AC}$$

$$\Rightarrow BA = 9 \text{ m and } AC = 9\sqrt{3} \text{ m}$$

$$\therefore BC = \sqrt{AC^2 - AB^2} = \sqrt{243 - 81} = \sqrt{162} = 9\sqrt{2} \text{ m}$$

$$\text{Hence, area of the field} = AB \times BC = 9 \times 9\sqrt{2} \text{ m}^2 = 81\sqrt{2} \text{ m}^2$$



Ex.5 A vertical tower stands on a horizontal plane and is surmounted by a vertical flag staff of height h . At a point on the plane, the angle of elevation of the bottom and the top of the flag staff are α and β respectively. Prove that the height

of tower is $\frac{h \tan \alpha}{\tan \beta - \tan \alpha}$.

Sol. Let AB be the tower and BC be the flag staff. Let O be a point on the plane containing the foot of the tower such that the angles of elevation of the bottom B and top C of the flagstaff at O are α and β respectively. Let $OA = x$ metres, $AB = y$ metres and $BC = h$ metres.

In $\triangle OAB$, we have

$$\cot \alpha = \frac{OA}{AB} \Rightarrow \cot \alpha = \frac{x}{y} \Rightarrow x = y \cot \alpha \quad \dots(i)$$

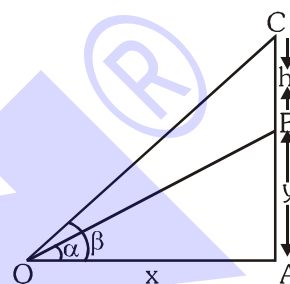
In $\triangle OAC$, we have

$$\cot \beta = \frac{x}{y+h} \Rightarrow x = (y+h) \cot \beta \quad \dots(ii)$$

Equating the values of x from (i) and (ii), we get

$$\begin{aligned} y \cot \alpha &= (y+h) \cot \beta \\ \Rightarrow y \cot \alpha - y \cot \beta &= h \cot \beta \\ \Rightarrow y (\cot \alpha - \cot \beta) &= h \cot \beta \end{aligned}$$

$$\Rightarrow y = \frac{h \cot \beta}{\cot \alpha - \cot \beta} \Rightarrow y = \frac{h / \tan \beta}{\frac{1}{\tan \alpha} - \frac{1}{\tan \beta}} = \frac{h \tan \alpha}{\tan \beta - \tan \alpha}$$



Ex.6 A spherical ball of diameter δ subtends an angle α at the eye of an observer when the elevation of its centre

is β . Prove that the height of the centre of the ball is $\frac{1}{2} \delta \sin \beta \operatorname{cosec} \left(\frac{\alpha}{2} \right)$.

Sol. O is the position of eye.

As is clear from figure, from $\triangle ODC$,

$$OC = \frac{h}{\sin \beta}$$

From $\triangle OAC$,

$$\sin \frac{\alpha}{2} = \frac{CA}{OC} = \frac{\frac{\delta}{2}}{h / \sin \beta}$$

$$\Rightarrow h = \frac{1}{2} \delta \sin \beta \cdot \operatorname{cosec} \frac{\alpha}{2}.$$

